Rayaan Juvale

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Education

University of Massachusetts Amherst — Amherst, MA
Master of Science in Data Science — *May 2026*
Relevant Coursework: Machine Learning, Data Engineering, Distributed Systems, Quantitative Analysis

University of Mumbai — Mumbai, IN
Bachelor of Engineering in Electronics and Telecommunication — *May 2024*
Relevant Coursework: Neural Networks, Object Oriented Programming, Python Programming

Work Experience

University of Massachusetts Amherst — Amherst, MA
Data Intern — *Sep 2024 – Present*

- Analyzed 40+ weeks of food waste data using Python and SQL, surfacing weekly trends to support sustainability decisions at UMass Dining.
- Tracked $500K+ in annual purchases using FoodPro, identifying cost-saving patterns from large-scale invoice data.
- Designed Tableau dashboards visualizing waste and purchasing patterns, reducing excess inventory costs.

HDFC ERGO General Insurance Company — Mumbai, IN
Software Developer Intern — *Aug 2023 – Apr 2024*

- Developed web applications using **JavaScript, HTML, CSS, and C#** in Microsoft's **ASP.NET MVC** framework, improving page load speed by 6.3%.
- Optimized **distributed database queries** over large-scale confidential datasets, boosting retrieval efficiency by 9%.
- Incrementally developed and tested the Ezeepay app cross-functionally, integrating user feedback to enhance functionality and reliability.

Clairvoyant Software Technologies Pvt. Ltd. — Mumbai, IN
Data Science Intern — *Jun 2023 – Aug 2023*

- Improved **information retrieval** by optimizing **MySQL** queries, cutting execution time by 13% on large-scale datasets.
- Implemented **predictive ML models** using Python and Scikit-learn, improving forecasting accuracy by 3.2%.
- Developed advanced data visualizations to enhance stakeholder insights and data-driven decision-making.

Project Experience

Real-time Anomaly Detection Engine | *Python, PyTorch, Kafka, Isolation Forest, FastAPI, MLflow, GCP*

- Architected a **real-time anomaly detection system** ingesting live data streams via **Apache Kafka**, running an **ensemble of 3 ML models** in parallel (Isolation Forest, LSTM Autoencoder in PyTorch, and Z-score control charts) each capturing different anomaly types across **10K+ events/sec**.
- Built a **real-time feature pipeline** computing rolling statistics, lag features, and rate-of-change signals with a **weighted ensemble voting layer** that combines all 3 model outputs into a single confidence score, cutting false positive rate by **41%** compared to any single model alone.
- Shipped a full-stack **live monitoring dashboard** using **React, D3.js, and WebSockets** on **GCP Cloud Run** with **MLflow** tracking model versions and experiments across the full development lifecycle.

German Credit Risk Classification | *Python, XGBoost, Scikit-learn, GridSearchCV*

- Built an end-to-end supervised **ML pipeline** across **1,000 records and 20 features**, benchmarking Logistic Regression, KNN, Random Forest, and **XGBoost** using **5-fold cross-validated GridSearchCV** optimized on F1 score.
- Selected **XGBoost** as the best performing model hitting a test **F1 of 0.5946** and outperforming all baselines by up to **14%** through systematic hyperparameter tuning.
- Preprocessed imbalanced data using **IQR-based outlier capping**, **OneHotEncoding**, and **StandardScaler** inside `sklearn` Pipelines to prevent data leakage throughout training and evaluation.

Reddit Data Engineering and Analysis | *Python, PostgreSQL, Docker, Streamlit, ETL*

- Designed a **distributed ETL pipeline** that processed **100K+ Reddit records** by combining web crawling with **PostgreSQL** for large-scale storage and fast retrieval.
- Containerized the entire pipeline with **Docker** so it runs consistently across any environment without manual setup.
- Built an **interactive Streamlit dashboard** showing real-time subreddit trends across high-volume data streams.

Technical Skills

Languages & Databases: Python, SQL, JavaScript, C#, Java, PostgreSQL, NoSQL, Node.js
Machine Learning & AI: PyTorch, TensorFlow, Scikit-learn, XGBoost, LSTM, Ensemble Methods, NLP, Generative AI, MLflow
Systems & Backend: Distributed Systems, REST APIs, WebSockets, Microservices, FastAPI, Redis, Apache Kafka
Cloud & DevOps: GCP (Cloud Run, GKE, Cloud Functions, Vertex AI, BigQuery), AWS (SageMaker, Lambda, S3, EC2), Docker, Kubernetes, Terraform, CI/CD, Linux/Unix
Data Engineering: Apache Spark, Airflow, Snowflake, BigQuery, Databricks, D3.js